Wildfire History Atlas

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Introduction

Historical wildfire datasets contain diverse dimensions, including geographical location, temporal positioning, event magnitude, and environmental context. Researchers investigating patterns may therefore need to examine multiple perspectives simultaneously; however, traditional static visualizations are limited in their ability to communicate and connect these dimensions within a single analytical workflow.

RQ1: How can multiple coordinated visualizations improve exploratory analysis of historical wildfire occurrence?

Wildfire History Atlas (Figure 1) was developed as an interactive exploratory tool that integrates complementary visualizations of historical wildfire data within a centralized interface. By connecting spatial, temporal, environmental, and quantitative perspectives, the application enables researchers to investigate wildfire occurrence from multiple analytical viewpoints.

Dataset

Wildfire History Atlas utilizes the Monitoring Trends in Burn Severity (MTBS) dataset, which is constructed by an interagency program aiming to map the extent and burn severity of large fires throughout the United States from 1984 to present. This project builds on this extensive historical dataset by providing an interactive environment for exploratory analysis of wildfire occurrence and spatial patterns.

Reference: Eidenshink, J., B. Schwind, K. Brewer, Z. Zhu, B. Quayle, and S. Howard. 2007. A project for monitoring trends in burn severity. Fire Ecology 3(1): 3–21.

Table 1. Primary MTBS attributes used by Wildfire History Atlas.

Column Name	Importance
`BurnBndAc`	Area burned by each fire, in acres
`Ig_Date`	Date of fire ignition
`geometry`	Spatial polygon representing the fire perimeter

Spatial Patterns

Where Have Fires Occurred?

Choropleths are thematic geospatial maps that use color gradations or patterns to represent aggregated quantitative data across predefined geographic units. This makes them useful for examining the spatial distributions of wildfire activity.

In Wildfire History Atlas, each geographic unit is a subregion; the subregions for the national map are states, and for the state-level maps, they are counties. The metric—either fire count or area burned—is aggregated for each subregion and represented using a color scale, enabling efficient identification of geographic patterns (Figure 2).

Because wildfire activity is not distributed uniformly across geographic regions, different color-encoding strategies may emphasize different aspects of the underlying distribution. Wildfire History Atlas therefore provides four choropleth encoding methods, summarized in Table 2.

Table 2. Choropleth encoding methods and their analytical purposes.

Choropleth Type	Explanation	Analytical Purpose
Regular Choropleth	Maps raw values using a continuous color scale	Preserving absolute differences between subregions
Quantile Choropleth	Divides subregions into five groups based on rank	Highlighting relative position and regional differences
Log-Scaled Choropleth	Applies a base-10 logarithmic transformation before color mapping	Revealing local variation across large ranges
Clipped Choropleth	Limits the influence of extreme values before color mapping	Improving visual contrast when outliers dominate the distribution

Temporal Patterns

How Has Fire Activity Changed Over Time?

Complementing the spatial perspective provided by choropleth maps, histograms allow researchers to examine temporal patterns in wildfire activity. For a selected region, Wildfire History Atlas aggregates wildfire occurrences by year; each bar represents the number of fires recorded that year. This provides a direct view of changes in fire activity over the available historical record (Figure 2).

The temporal extent of each histogram begins with the first year containing recorded fire data and ends with the most recent year containing data. Years without recorded data are retained as empty intervals within this range rather than interpreted as years with zero fires. This distinction preserves the limitations of the underlying dataset while allowing users to identify periods of increased or decreased recorded fire activity without assuming wildfire absence.

Together, the spatial and temporal visualizations provide complementary perspectives on wildfire occurrence: **choropleths identify where fire activity is concentrated, while histograms reveal how recorded activity varies over time.**



Figure 1: Wildfire History Atlas application interface integrating spatial, temporal, contextual, and quantitative perspectives.

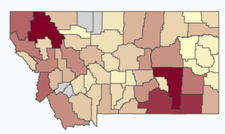


Figure 2: Choropleth of Montana.

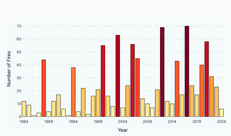


Figure 3: Histogram for Montana.

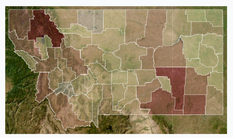


Figure 4: Montana satellite imagery.

Contextual Patterns

Which Environmental Variables Surround Wildfire Occurrence?

To facilitate examination of the spatial relationship between wildfire occurrence and environmental variables, the system architecture supports multi-layered geographic visualization. **Viewing a basemap underneath the choropleth polygons allows researchers to compare the distribution of wildfire activity with underlying geographic characteristics of each region.** A variety of basemaps are available, and the analytical utility of each basemap layer is summarized in Table 3:

Table 3. Basemap layers and their analytical purposes.

Basemap	Analytical Purpose
Satellite	Comparing wildfire patterns with vegetation and land cover
Streets	Revealing roads and populated places within a region
Topographic	Highlighting mountain ranges, rivers, and elevation
Light Gray	Minimizing visual distraction and emphasizing the data
Earth at Night	Comparing wildfire patterns with nighttime light density

After selecting a basemap, the user has the option to change the opacity of the choropleth polygons to prevent visual occlusion and balance the visibility of the layers. Each basemap is assigned a recommended default opacity setting optimized for visual readability, but users can adjust the opacity setting according to their analytical needs or preference.

Satellite Mode (Figure 4) is currently implemented as a prototype and does not yet support all interactions available in the standard choropleth view.

Quantitative Summaries

What Additional Information Can Be Explored?

Once users identify a region or subregion of interest, **the statistics panel provides additional quantitative context complementary to the spatial and temporal visualizations.**

When a user selects a subregion on a choropleth map, the statistics panel displays five quantitative values:

- Selected metric: fire count or area burned
- Percent of total: proportion represented by the selected metric
- Additional context: the value of the metric not selected for visualization
- Percent of total: proportion represented by the additional metric
- Average area burned per fire

For histograms, users can display a mean line to identify years with above- or below-average recorded fire activity. The mean is calculated using only years with recorded wildfire data; years with missing data are excluded rather than interpreted as zero-fire years.

Conclusion

Wildfire History Atlas provides an integrated environment for exploring multiple dimensions of historical wildfire occurrence within MTBS data.

Choropleth maps reveal spatial patterns in wildfire activity and enable efficient pattern recognition through multiple approaches to geographic comparison. Histograms provide a complementary perspective of temporal patterns, while basemap overlays provide environmental context for spatial observations. The statistics panel provides additional quantitative context for the current visualization.

By connecting these perspectives within a single interactive application, Wildfire History Atlas provides a flexible exploratory tool for researchers investigating where, when, and in what context wildfire activity occurs.

Future Work

Future development of Wildfire History Atlas will expand the application's geographic coverage, analytical capabilities, and interactive functionality.

Planned directions include:

- Ecoregion support and other projections
- Additional metrics and statistical analysis
- Normalization

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